

Immunoglobulin-Mediated Glomerulonephritis With a Membranoproliferative Pattern of Injury and Cryoglobulinemic Glomerulonephritis

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IMMUNOGLOBULIN-MEDIATED GLOMERULONEPHRITIS WITH A MEMBRANOPROLIFERATIVE PATTERN OF INJURY

Definition and Classification

Membranoproliferative glomerulonephritis (MPGN) is a pattern of glomerular injury resulting from subendothelial and mesangial deposition of immune complexes and/or complement factors and their products, along with proliferative changes in the glomeruli. This pattern does not represent a disease entity per se. We therefore prefer to use terminology that refers to the underlying disease, such as immune-complex glomerulonephritis, monoclonal Ig glomerulonephritis, and C3 glomerulopathy.¹ Other terms, such as lobular glomerulonephritis and mesangiocapillary glomerulonephritis, should be avoided.

Former Classification

Previously, MPGN was classified as MPGN type I, MPGN type II, and MPGN type III based on ultrastructural characteristics. MPGN type I was defined by the presence of mesangial and subendothelial deposits. MPGN type II, also referred to as *dense deposit disease* (DDD), was characterized by mesangial and intramembranous highly electron-dense deposits. MPGN type III was defined by the presence of subendothelial, intramembranous, and subepithelial electron-dense deposits. However, some cases had glomerular deposition of immunoglobulins and complement components, whereas in others, only complement deposition was present. This insight engendered a new classification. Older publications regarding epidemiology, therapy, and transplantation may be difficult or impossible to interpret because, with the exception of DDD, immune-complex and complement-mediated MPGN were included together (see later discussion).

Newer Classification Based on Etiology and Pathogenesis

The Mayo Clinic classification of MPGN^{2,3} divides MPGN based on two broad pathogenetic pathways: (1) immune complex or monoclonal immunoglobulin deposition in the glomeruli with or without complement deposition and (2) complement deposition subsequent to dysregulation of the complement system.

The MPGN pattern of injury secondary to deposition of immune complexes in the glomeruli can occur when there are persistent circulating antigen-antibody immune complexes resulting from chronic infections or autoimmune diseases. MPGN with immunoglobulin and complement deposits may also be observed in the setting of a monoclonal gammopathy.

On light microscopy, a proliferative glomerulonephritis is seen in the acute phase with initial influx of neutrophils followed by mononuclear inflammatory cells. In the repair phase, the injured mesangial cells and endothelium produce new basement membrane, and there is expansion of the mesangial matrix. As a result of remodeling, the glomerular basement membrane (GBM) thickens and glomeruli acquire a lobular appearance typical of MPGN pattern. The immune-complex and monoclonal immunoglobulin-mediated MPGN are differentiated by the presence of immunoglobulins on immunofluorescence (IF); in the former the deposits are polyclonal, whereas in the latter they are monoclonal. C3 is also usually present, indicating activation of the complement pathway.

Complement-mediated MPGN is less common than immune complex-mediated MPGN and results from dysregulation of the complement alternative pathway (see [Chapter 23](#)).

A third pathogenic pathway with an MPGN pattern despite the absence of immune complex or complement deposition can occur with chronic thrombotic microangiopathy or chronic endothelial injury ([Table 22.1](#)).

Epidemiology

MPGN is decreasing in frequency in high-income countries (possibly because of better control of infectious diseases) but remains common in patients presenting with nephrotic syndrome in low- and middle-income countries in South America, Asia, and Africa. The incidence of MPGN in most high-income countries is less than 1 to 2 cases per million population per year.⁴ Only about 4% to 5% of kidney biopsy samples showing an MPGN pattern have underlying C3 glomerulonephritis (see [Chapter 23](#)).

Clinical Presentation and Pathology

Infection-Associated MPGN

Chronic viral infections such as hepatitis C (HCV) and B (HBV) are important causes of immune complex-mediated MPGN (cryoglobulinemic glomerulonephritis; see later discussion and [Chapter 57](#)).^{5,6} The exact mechanism of HCV-related glomerular disease is unknown. Toll-like receptors, particularly TLR3, might play a role in HCV-related MPGN.⁷ HBV most frequently causes a membranous nephropathy pattern of injury but can more rarely be associated with MPGN. The pathology of HBV-related MPGN is likely related to the trapping of circulating immune complexes in the mesangium and subendothelial regions.⁸ It is also possible that mesangial cells are directly infected by HBV virions. Uncommonly, MPGN may be seen with acute viral infections, such as with Puumala hantavirus.⁹

Chronic bacterial infections can cause MPGN as a result of continuous low-grade antigenemia.^{10,11} Potential causative agents include *Staphylococcus*, *Mycobacterium tuberculosis*, *Streptococcus*,

TABLE 22.1 Causes of a Membranoproliferative Pattern of Glomerular Injury

Immune complex mediated, monoclonal immunoglobulin mediated	Deposition of immune complexes as a result of an infection	Viral: hepatitis B and C Bacterial: endocarditis, infected ventriculoatrial shunt, visceral abscesses, leprosy, meningococcal meningitis Protozoa/other infections: malaria, schistosomiasis, mycoplasma, leishmaniasis
	Deposition of immune complexes as a result of an autoimmune disease	Systemic lupus erythematosus Sjögren syndrome Rheumatoid arthritis
	Deposition of monoclonal immunoglobulin as a result of a monoclonal gammopathy	Plasma cell or B-cell disorder
Complement-mediated (C3 glomerulonephritis and dense deposit disease)	Mutations in complement-regulating proteins CFH, CFI, CFHR5	
	Antibodies to complement regulating proteins C3/C4 nephritic factor, antibodies against CFH, CFI, or CFB	
Non-immunoglobulin mediated, non-complement mediated	Mutations in complement factors C3, CFB	
	Monoclonal gammopathy	
Idiopathic	Healing phase of HUS/TTP	
	Antiphospholipid (anticardiolipin) antibodies syndrome	
	POEMS syndrome	
	Radiation nephritis	
	Nephropathy associated with bone marrow transplantation	
	Drug-associated thrombotic microangiopathies	
	Sickle cell anemia and polycythemia	
	Dysfibrinogenemia and other prothrombotic states	
	Transplant glomerulopathy	
Idiopathic	None of the conditions previously mentioned present	

CFB, complement factor B; CFH, complement factor H; CFI, complement factor I; CFHR5, complement factor H related 5; HUS, Hemolytic uremic syndrome; POEMS, polyneuropathy, organomegaly, endocrinopathy, monoclonal gammopathy, skin; TTP, thrombotic thrombocytopenic purpura. Modified from Fervenza FC, Sethi S, Glassock RJ. Idiopathic membranoproliferative glomerulonephritis: Does it exist? *Nephrol Dial Transplant*. 2012;27(12):4288–4294.

Propionibacterium acnes, *Mycoplasma pneumoniae*, *Coxiella burnetii*, *Nocardia*, *Brucella*, and *Meningococcus*.^{12,13} So-called “shunt nephritis,” as may be seen with ventriculoatrial or ventriculocaval shunts for hydrocephalus or as a result of catheter infections (e.g., parenteral nutrition), is caused most commonly by coagulase-negative staphylococci. Fungal and parasitic infections less frequently cause an MPGN pattern of injury.^{14,15}

In the setting of viral infections, IF typically shows granular deposition of immunoglobulin M (IgM), C3, and both kappa and lambda light chains (Fig. 22.1). IgG may or may not be present, and C1q is typically negative. The presence of IgG staining stronger than IgM and C3 is more often seen in bacterial infections, although in some cases with chronic infections the immunoglobulin staining may be weak compared with the C3 staining. Electron microscopy (EM) shows mesangial and subendothelial deposits. The capillary walls are thickened with entrapment of cellular elements, subendothelial electron-dense deposits, matrix-like material, and new GBM formation manifested as double contours on light microscopy.

Additional information on the glomerular diseases associated with infection can be found in Chapter 57.

Autoimmune-Associated MPGN

An MPGN pattern of injury is commonly seen in patients with autoimmune diseases such as systemic lupus erythematosus (SLE), rheumatoid arthritis (RA), primary Sjögren syndrome, undifferentiated connective tissue disease, primary sclerosing cholangitis, and Graves disease.^{16,17} The majority of patients with Sjögren syndrome and glomerulonephritis (80%) have type II monoclonal cryoglobulinemia and low complement C4 levels.¹⁸ Sjögren syndrome is one of the most common causes of non-HCV-related cryoglobulinemia.¹⁹

On IF, a full house pattern with positive staining for IgG, IgA, IgM, C1q, C3, and kappa and lambda light chains is frequently seen, particularly in the setting of SLE. IgM may be the dominant immunoglobulin in MPGN associated with RA and primary Sjögren syndrome. EM shows mesangial and subendothelial capillary wall electron-dense deposits. Subepithelial deposits may also be present. In such cases, an exostosin 1/exostosin 2-associated membranous nephropathy should be considered.²⁰ Tubuloreticular inclusions are often present in the endothelial cells.

Monoclonal Immunoglobulin–Associated MPGN

Monoclonal gammopathies encompass disorders characterized by clonal proliferation of immunoglobulin-producing B lymphocytes or plasma cells. In most patients with monoclonal immunoglobulin-associated MPGN, a monoclonal immunoglobulin cannot be detected in the blood or urine^{21,22} with routine serum electrophoresis and immunofixation examination, and evidence of overt malignancies such as lymphoma, multiple myeloma, or Waldenström macroglobulinemia is absent. In those cases, the term *monoclonal gammopathy of renal significance* (MGRS) is used.²³ The glomerular diseases included in this group are proliferative glomerulonephritis with monoclonal immunoglobulin deposits (PGNMID), amyloidosis, fibrillary glomerulonephritis (DNAJB9 positive), immunotactoid glomerulopathy, and monoclonal immunoglobulin deposition disease (see Chapter 28).^{24–26} Although each of these entities may have a varied morphologic appearance, the common pattern of glomerular injury is MPGN.

On kidney biopsy (Fig. 22.2), glomerular deposition of the monoclonal immunoglobulin usually results in an MPGN pattern. Less common patterns of proliferative glomerulonephritis include mesangial proliferative, diffuse proliferative, crescentic, necrotizing,

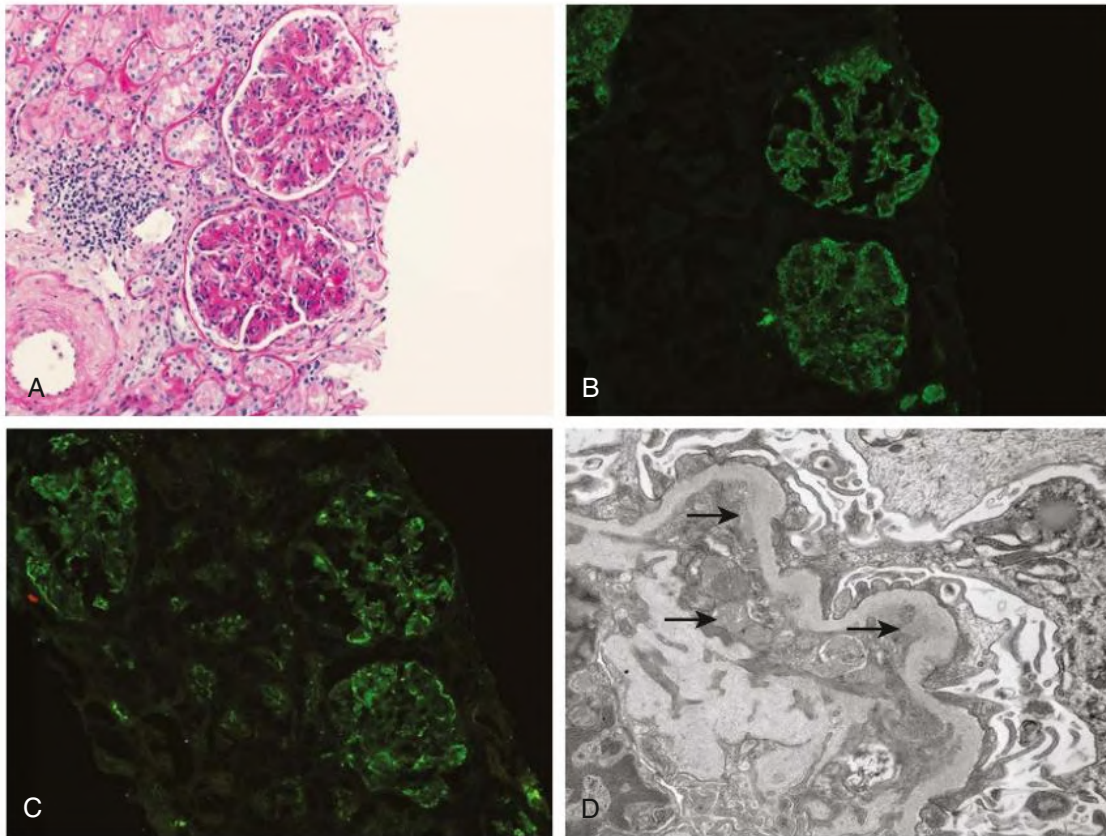


Fig. 22.1 Membranoproliferative Glomerulonephritis (MPGN) Associated With Hepatitis C. (A) Light microscopy showing two glomeruli with an MPGN pattern of injury (PAS $\times 20$). (B–C) Immunofluorescence microscopy showing bright granular staining for immunoglobulin M (B), and C3 (C). There was equal staining for kappa and lambda light chains (not shown). (D) Electron microscopy showing subendothelial electron-dense deposits and double contour formation. *Black arrows* point to subendothelial deposits. (D, $\times 13,000$.)

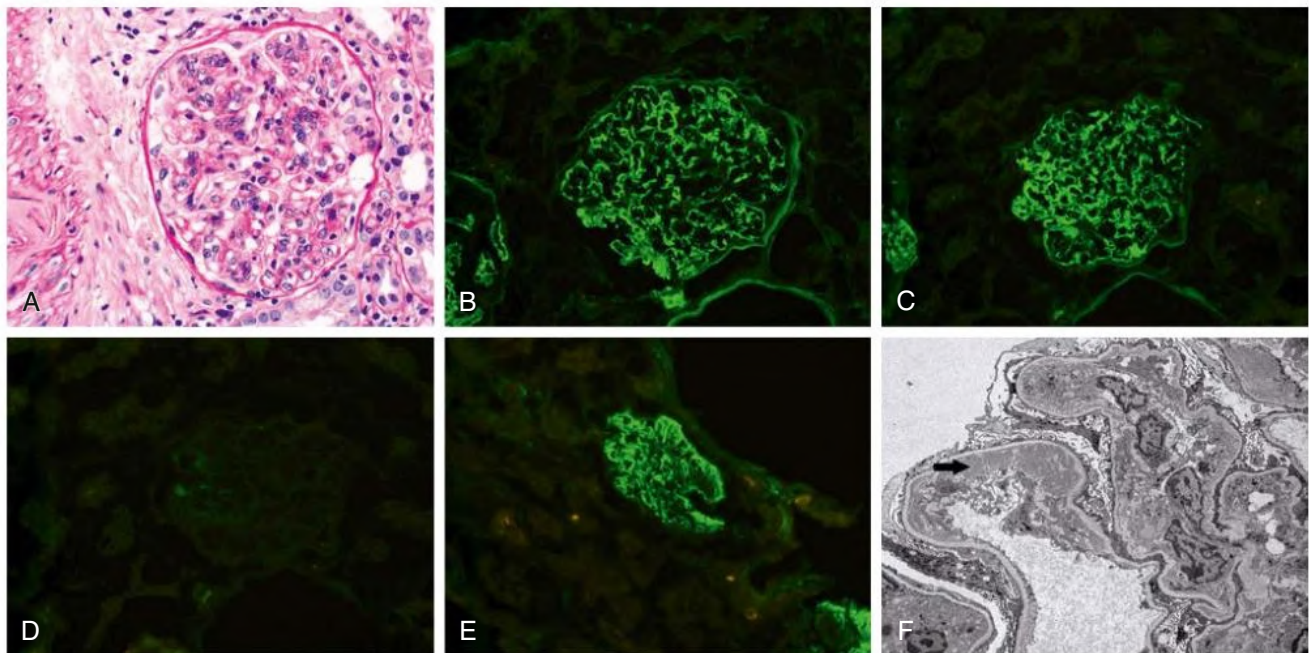


Fig. 22.2 Proliferative Glomerulonephritis With Monoclonal Immunoglobulin Deposits. (A) Light microscopy showing a membranoproliferative glomerulonephritis (MPGN) pattern of injury (PAS, $\times 40$). (B–E) Immunofluorescence microscopy showing bright granular staining for immunoglobulin G (IgG) (B), kappa light chains (C), negative staining for lambda light chains (D), and positive staining for IgG3 subtype (E). (F) Electron microscopy showing subendothelial electron-dense deposits and double contour formation. *Black arrows* point to subendothelial deposits (F, $\times 4000$.)

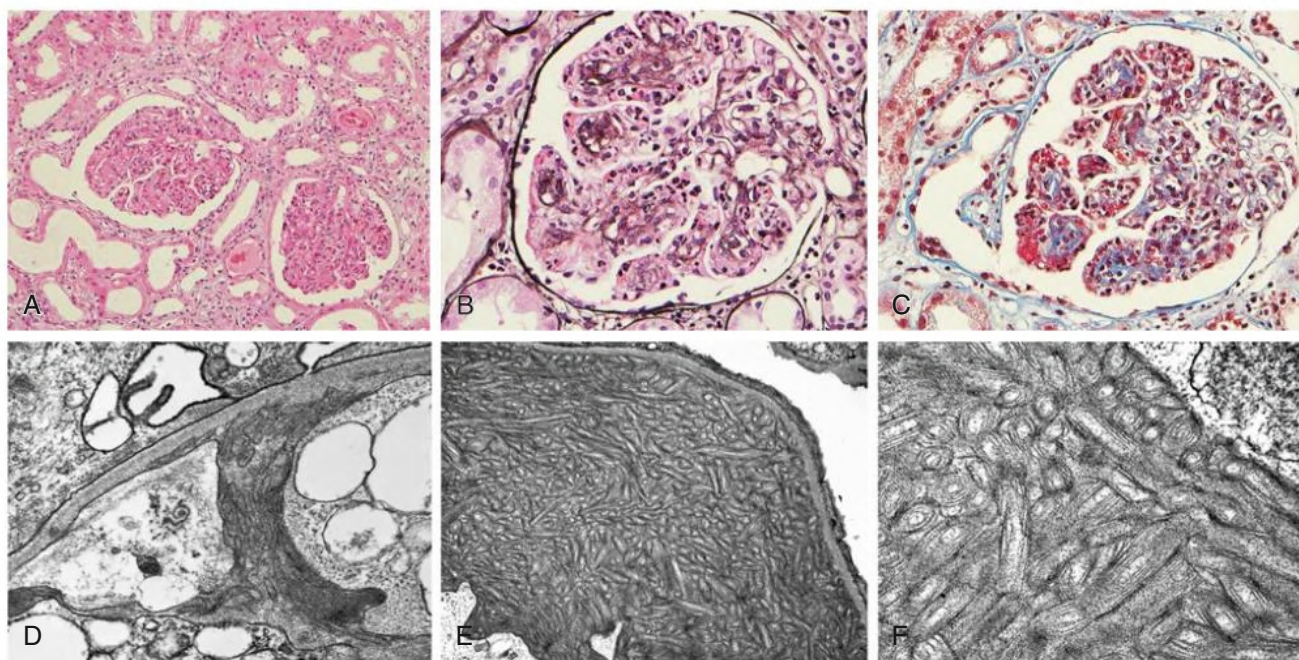


Fig. 22.3 Cryofibrinogen Glomerulopathy With a Membranoproliferative Pattern of Injury. (A–C) Light microscopy showing a membranoproliferative glomerulonephritis pattern of injury. Scattered fuschinophilic material can be seen with some capillary loops in part C. Immunofluorescence microscopy was negative for immune deposits. (D–F) Electron microscopy showing randomly arranged and loosely arranged subendothelial fibrillary deposits with tubular substructures (D) and aggregates of intraluminal tubular deposits with large central bore with double and triple layering (E–F). (A, H&E, $\times 20$. B, Silver methenamine silver, $\times 40$. C, Masson trichrome stain, $\times 40$. D, $\times 30,000$. E, $\times 9300$. F, $\times 3000$.)

and sclerosing glomerulonephritis.^{21,22} IF studies are crucial for the diagnosis and show mesangial and capillary wall monoclonal immunoglobulin deposits. The monoclonal immunoglobulin most often contains heavy chain IgG, less commonly IgM or, rarely, IgA, with kappa or lambda light chain restriction. Sometimes, only heavy or light chains may be present.²⁷ EM shows mesangial and subendothelial electron-dense deposits, and, rarely, subepithelial and intramembranous deposits. Glomerular capillary wall remodeling with double contour formation is often present.

In cases in which the heavy chain consists of IgG, the IgG3 subclass is the most common subclass. Interestingly, this class of deposits is most likely to have undetectable circulating monoclonal immunoglobulin by routine serum and urine electrophoresis studies.²⁸

In monoclonal immunoglobulin-associated MPGN, 30% of patients have a monoclonal or biclonal band on serum immunofixation studies. Of these, bone marrow studies revealed 16 cases of MGRS, of which two converted to multiple myeloma; two cases of chronic lymphocytic leukemia (CLL); one case of lymphoplasmacytic lymphoma/Waldenström macroglobulinemia; three cases of low-grade B-cell lymphoma not further classifiable; and six cases of multiple myeloma.²⁵ Thus, all patients with MPGN associated with monoclonal immunoglobulin should be evaluated for an underlying plasma cell or B-cell proliferative disorder.

MPGN With Masked Immune Deposits

MPGN with isolated C3 deposits rarely occurs in the setting of monoclonal gammopathy.²⁹ Hypocomplementemia is common. IF on formalin-fixed, paraffin-embedded tissue after protease digestion unmasks monoclonal immunoglobulin glomerular deposits. The immunoglobulin on the glomerular deposits matches the monoclonal protein on serum immunofixation. It is important that these cases are not misdiagnosed as C3 glomerulopathy because most are associated

with a low-grade lymphoma or plasma cell dyscrasia. A C4d stain can be helpful in detecting masked immune deposits.^{30,31}

Complement-Mediated MPGN

Complement-mediated MPGN is also the most common MPGN pattern seen in C3 glomerulopathy, a lesion that is characterized by dominant C3 staining with minimal or no immunoglobulin deposits (see Chapter 23).

MPGN Without Immunoglobulins or Complement

Glomeruli may demonstrate membranoproliferative-like changes in the absence of immunoglobulin or complement deposits in chronic thrombotic microangiopathy (see Chapter 30).

Uncommon Causes of MPGN

Cryofibrinogen-related MPGN. Cryofibrinogenemia is a rare disease resulting in an MPGN pattern of injury (Fig. 22.3). Cryofibrinogen is a cryoprecipitate that develops after refrigeration of plasma but not serum (when both serum and plasma form a precipitate on refrigeration, the responsible proteins are called cryoglobulins). Cryofibrinogenemia may be asymptomatic but can be associated with thromboembolic disease, particularly affecting the skin.

C4 glomerulopathy. C4 glomerulopathy is characterized by glomerular deposits of predominantly C4 with little or no immunoglobulin or C3 deposition. This glomerulopathy encompasses C4 DDD and C4 glomerulonephritis³² and may be caused by an overactive lectin pathway of complement. Kidney biopsy shows a membranoproliferative pattern of injury with extremely thick glomerular capillary walls.

Collagen type III glomerulopathy. Collagen type III (collagenofibrotic) glomerulopathy is a rare disorder characterized by massive accumulation of atypical type III collagen fibrils in the mesangium and subendothelial space (see Chapter 29).

BOX 22.1 Patient Evaluation in Cases of MPGN Pattern of Injury on Biopsy^a

Immune Complex MPGN

Infection

- Viral serology for HBV, HCV, HIV, COVID-19
- Quantification of viral load
- Blood cultures
- Imaging studies for deep-seated abscesses and vegetations by echocardiogram
- QuantiFERON-Tb Gold
- Parasitic and fungal infections are investigated only in the appropriate clinical situation (history of recent travel to endemic regions, prolonged fever of unknown origin, atypical pulmonary infiltrates)

Autoimmune

- Antinuclear antibodies, anti-ds-DNA antibodies, extractable nuclear antigen antibodies (anti-ribonucleoprotein, anti-SSA, [Ro], anti-SSB, [La], anti-Smith [Sm], Scl-70, Anti-Jo-1)

Monoclonal Immunoglobulin-Associated MPGN

Serum and urine protein electrophoresis and immunofixation

Serum M-protein isotype by mass spectrometry

Serum free light chains

Bone marrow biopsy

MPGN Without Immunoglobulin and Complement Deposition

Evaluate for thrombotic microangiopathy

^aC3/C4, cryoglobulins, rheumatoid factor, C-reactive protein, serum albumin, urinalysis, quantified proteinuria and kidney function should be performed at baseline in all patients.

HBV, Hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; MPGN, membranoproliferative glomerulonephritis.

Evaluation

The evaluation for MPGN should be guided by the suspected cause, based on clinical presentation and biopsy findings. A standard workup for MPGN is presented in Box 22.1. Low serum C3 and C4 are more typical of classic pathway activation, as in immune complex-mediated glomerulonephritis. A low C3 with a normal C4 suggests abnormalities of the alternative complement pathway. These findings, along with the biopsy results (particularly IF and EM), can guide further detailed evaluation.

Infections

Workup in the evaluation of infection-related MPGN depends on the suspected pathogen (see Chapter 57). In the case of viral infection-related MPGN, tests should include viral serology and quantification of viral load. The workup for parasitic infection-related MPGN should include blood tests for malaria, urine and stool tests for schistosomiasis, and serologic tests for schistosomiasis and leishmaniasis. Blood cultures, cultures of indwelling catheter tips, imaging studies for deep-seated abscesses, and transthoracic echocardiograms for valvular vegetations should be performed in the case of suspected fungal and bacterial infections.

Autoimmune Diseases

The diagnostic tests for autoimmune diseases should follow the established criteria, such as those of the American College of Rheumatology.

Monoclonal Gammopathy

The workup should include serum protein electrophoresis, urine protein electrophoresis, serum and urine immunofixation, and serum free

light chain assays. Urine free light chains have no diagnostic utility and should not be assessed. Bone marrow evaluation should be performed to rule out an underlying plasma cell dyscrasia and/or lymphoproliferative disorder. In patients in whom a clone is identified (approximately 20%–30%), the underlying clonal process is plasmacytic in 50% of patients and derives from B-cell lymphocytes in the other 50%.³³ In our experience, in the absence of a circulating monoclonal protein, the probability of finding an underlying plasma cell dyscrasia is limited.²⁸

Treatment

Therapy should be directed to treatment of the underlying condition, such as chronic infection, autoimmune disease, and malignancy (e.g., myeloma, lymphoma, CLL). Treatment of monoclonal immunoglobulin-associated MPGN is complex and depends on the type of the immunoglobulin. In patients with IgM monoclonal proteins, we suggest employing a regimen used to treat Waldenström macroglobulinemia as initial therapy: rituximab 4 doses of 375 mg/m², weekly for 4 weeks, with an identical regimen repeated 6 months later. More intense regimens include bendamustine plus rituximab; bortezomib, dexamethasone, and rituximab; or bortezomib plus rituximab with or without dexamethasone.

For patients with non-IgM monoclonal proteins, conservative and immunosuppressive therapy with the use of corticosteroids (alone or in combination with an alkylating agent), thalidomide, and mycophenolate mofetil (MMF) have been used in a small number of patients with variable outcomes.³⁴ A recent case series suggests that clone-directed therapy with the use of rituximab; bortezomib plus dexamethasone; or bortezomib, cyclophosphamide, and dexamethasone, similar to that used in myeloma, results in a good chance of partial or complete proteinuria remission.³³ In nontransplant patients who were treated empirically with these regimens, the rate of complete or partial remission of proteinuria was 75%. These regimens are well tolerated and safe for use in kidney failure, with minimal dose adjustments needed. We typically treat patients for at least 6 months and then reassess. Some patients may also respond with disappearance of the monoclonal gammopathy after a 3- to 6-month combined course of prednisone and cyclophosphamide.

Daratumumab is an IgG1-kappa human monoclonal antibody that binds to the CD38 transmembrane glycoprotein on the surface of tumor cells and induces apoptosis. It has shown promise in the treatment of refractory multiple myeloma. In a phase II trial in patients with proliferative glomerulonephritis and monoclonal immune deposits (PGNMID), the overall response rate at 12 months was 100% (60% partial and 40% complete remission of proteinuria).³⁵ Autologous stem cell transplantation has been reported to be successful in MGRS and light chain proximal tubulopathy, but little information is available on its use regarding monoclonal immunoglobulin-associated MPGN.³⁶

There is no strong evidence to guide treatment in patients with “idiopathic” MPGN.³⁷ Corticosteroids have been widely used as monotherapy and/or in combination with other immunosuppressives. In a randomized trial, 80 children with nephrotic-range proteinuria and preserved estimated glomerular filtration rate (eGFR) were assigned to either prednisone or lactose 40 mg/m² every other day for a mean duration of 41 months.³⁸ Prednisone therapy had a significantly lower rate of treatment failure (40% vs. 55%) but was associated with significant toxicity, particularly hypertension, and it is uncertain whether the findings (published in 1992) are applicable to contemporary practice. No randomized studies with corticosteroids have been performed in adults with idiopathic MPGN.

Cytotoxic agents have been tried, with unimpressive results and significant side effects. Calcineurin inhibitors have also been used in

TABLE 22.2 Cryoglobulin Classification and Associated Diseases

Cryoglobulin Type	Immunoglobulin Class	Associated Diseases
I. Monoclonal immunoglobulins	M > G > A > BJP	Myeloma, CLL, Waldenström macroglobulinemia
II. Mixed cryoglobulins with monoclonal immunoglobulins	M/G >> G/G	Infection (hepatitis C), Sjögren syndrome, CLL, lymphoma
III. Mixed polyclonal immunoglobulins	M/G	Infection (hepatitis B/C), SLE, RA, vasculitis, neoplasia

BJP, Bence Jones protein (light chains); CLL, chronic lymphocytic leukemia; RA, rheumatoid arthritis; SLE, systemic lupus erythematosus.

uncontrolled studies, suggesting a modest decline of proteinuria in patients with MPGN not otherwise characterized.

Data on the use of MMF in patients with MPGN are limited.^{39–41} An observational study compared five adults treated with MMF and oral prednisolone to six patients on no immunosuppressive agents.³⁹ At 18 months, proteinuria had declined and GFR was preserved or improved in the MMF group, compared with no change in proteinuria and a 40% decline in GFR in patients who were not treated. Whether the benefit was derived from MMF or prednisone cannot be ascertained from the study.

Our experience with rituximab (1 g on days 1 and 15) in four patients with idiopathic MPGN showed a 65% reduction in proteinuria and preserved GFR over 1 year.⁴²

The KDIGO 2021 Clinical Practice Guideline on Glomerular Diseases recognized difficulties in interpreting previous information because, in many instances, historical controls were used, studies antedated the use of angiotensin II blockade, statistical significance was marginal, or the power to detect substantial differences was small.³⁷ More importantly, in these studies, the underlying pathogenic processes that lead to MPGN were not taken into account. It also recognizes that truly “idiopathic” MPGN is an exceptional condition in adults. As such, KDIGO 2021 only provides practice points based on very low-quality evidence, clinical experience, and expert opinion. We have taken a pragmatic approach based on our clinical experience, as follows:

- Patients with normal kidney function, no active urinary sediment, and nonnephrotic range proteinuria can be treated conservatively as outlined in [Chapter 82](#) because the long-term outcome is relatively benign in this setting. Frequent follow-up is required initially to detect early deterioration in kidney function. If there is no disease progression, follow-up every 6 months should suffice.
- Patients with nephrotic syndrome and preserved kidney function could be treated with a regimen similar to that used in primary (permeability factor-related) focal segmental glomerulosclerosis, such as prednisone 1 mg/kg/day (maximum dose 60–80 mg/day) for 12 to 16 weeks (see [Chapter 19](#)). If the patient responds, prednisone is gradually tapered to alternate-day therapy over 6 to 8 months. If there is less than a 30% reduction in proteinuria after 12 to 16 weeks, we recommend tapering and discontinuing the prednisone. Calcineurin inhibitors may be considered in patients who do not respond to or tolerate glucocorticoids.
- Patients who present with impaired kidney function (eGFR < 60 mL/min/1.73m²), with or without nephrotic syndrome, but without crescents or severe tubulointerstitial fibrosis on kidney biopsy, and an active urinary sediment (arbitrarily defined as >10 red blood cells per high-power field) can also be considered for treatment with corticosteroids, as mentioned previously. For patients who cannot or do not wish to receive high-dose glucocorticoids, treatment with MMF, with or without low-dose prednisone (10 mg/day), for 6 to 12 months is a reasonable alternative. If there is no improvement in GFR, proteinuria, or hematuria after 6 to 12 months of therapy, we discontinue these agents and consider repeating a kidney biopsy to reevaluate disease activity and chronicity. If the repeat

kidney biopsy shows mainly chronic damage without mesangial proliferation and/or endocapillary hypercellularity, we do not give additional immunosuppressive therapy because it is unlikely to be beneficial. If the repeat kidney biopsy shows evidence of ongoing active glomerulonephritis, rituximab (1 g on day 1 and 1 g on day 15) or cyclophosphamide 2 mg/kg/day (reduced to 1.5 mg/kg/day in patients with a serum creatinine >2.5 mg/dL or age >60 years) for 3 to 6 months could be added, realizing that the efficacy of rituximab or cyclophosphamide in these patients is unknown. Patients who do not respond to treatment with either cyclophosphamide or rituximab are considered to have resistant disease and we recommend stopping immunosuppressive therapy, continuing general supportive measures, and referring them for kidney transplantation when appropriate.

- Patients presenting with rapidly progressive disease and crescents on biopsy can be treated as in other forms of crescentic glomerulonephritis with pulse methylprednisolone followed by oral corticosteroids and cyclophosphamide (see [Chapter 26](#)).
- Patients who present with eGFR less than 30 mL/min/1.73m² and severe tubulointerstitial fibrosis on kidney biopsy are unlikely to benefit from immunosuppressive therapy.

Transplantation

Recurrent MPGN after kidney transplantation is discussed in [Chapter 113](#).

CRYOGLOBULINEMIC GLOMERULONEPHRITIS

Definition

Cryoglobulins are immunoglobulins that precipitate at cold temperatures and dissolve when rewarmed. Cryoglobulins usually present in the setting of infections, autoimmune diseases, and monoclonal gammopathy, but, in rare instances, they can be idiopathic. All of the aforementioned causes can present with an MPGN pattern of injury with intraluminal deposits (immune microthrombi) suggestive of cryoglobulins.

Cryoglobulins are often divided into three types ([Table 22.2](#)). Type I cryoglobulinemia is characterized by the presence of a single monoclonal immunoglobulin, most often resulting from an underlying B-cell hematologic malignancy, typically Waldenström macroglobulinemia or multiple myeloma. In the absence of overt malignancy, the diagnosis of MGRS should be established. The cryoglobulins are usually IgM or IgG, but IgA and monoclonal free light chains also may occur.

In type II mixed cryoglobulinemia, there is a mixture of monoclonal IgM with rheumatoid factor activity directed against a polyclonal IgG. It is most commonly caused by chronic infections, generally HCV, but also HBV and HIV. It can also occur in the context of autoimmune diseases, particularly Sjögren syndrome. In patients with HCV infection, cryoglobulinemia may be the first manifestation of liver disease.^{6,43}

In type III mixed cryoglobulinemia, the IgG and IgM are polyclonal. Type III cryoglobulinemia is most commonly seen in HCV



Fig. 22.4 Purpura in a Patient with Hepatitis C Virus-associated Cryoglobulinemia. (A) Palpable nonpruritic purpura affecting the lower extremities in a patient with hepatitis C virus infection and cryoglobulinemia. Raised purpuric lesions are present on the legs. Differential diagnosis for patients with dermatorenal syndromes includes IgA vasculitis and antineutrophil cytoplasmic antibody-associated vasculitis. (B) Purpuric lesions are present on the patient's buttocks and thigh. Note the purpuric lesions along the superior and inferior elastic border of the undergarment line.

infection, in chronic inflammatory and autoimmune disease (e.g., SLE and Sjögren syndrome), and in lymphoproliferative malignancies. Thus, there is a significant overlap between the underlying causes of type II and III cryoglobulins.

Clinical Presentation

The syndrome of cryoglobulinemia is caused by deposition of antigen-antibody complexes in capillaries and small arterioles and occasionally in small arteries.⁴⁴ Most cases are associated with HCV. In the majority of patients with HCV infection, the immune complexes consist of HCV, anti-HCV IgG, and monoclonal IgM anti-IgG (with rheumatoid factor activity) (type II mixed cryoglobulinemia). Clinical presentation of mixed cryoglobulinemia includes weakness, palpable nonpruritic purpura (Fig. 22.4), symmetric arthralgias (arthritis is rare), peripheral neuropathy, and kidney involvement.⁴⁵ The purpura is usually painless and mainly localized to the extremities and buttocks, similar to patients with IgA vasculitis (see Chapter 24). Other manifestations include digital necrosis (Fig. 22.5), congestive heart failure, pulmonary infiltrates, and mesenteric ischemia. The clinical syndrome and dominant symptoms may evolve over time.

On the other hand, patients with type I cryoglobulinemia are usually asymptomatic. Kidney disease occurs in approximately 20% to 60% of the patients, ranging from microscopic hematuria with low-grade proteinuria to a full-blown nephrotic or nephritic syndrome or both. Hypertension is common and can be severe. Kidney disease is a major determinant of long-term prognosis and substantially increases mortality.

Evaluation

Laboratory data suggestive of a cryoglobulin are low C4 levels (and normal or mildly decreased C3 levels) and a positive rheumatoid factor. It is crucial that blood is drawn into collection tubes that have been prewarmed to 37°C without anticoagulants because anticoagulants



Fig. 22.5 Necrosis of the fingertips in a patient with cryoglobulinemia.

can produce false-positive results due to cryofibrinogen or heparin-precipitable complexes. However, in as many as 40% of patients with type II and in occasional patients with type III mixed cryoglobulinemia, circulating cryoglobulins may not be detected. In some cases, kidney disease presents in the absence of detectable cryoglobulins or other manifestations of mixed cryoglobulinemia such as purpura or arthritis. There is a poor correlation between the clinical manifestations and the cryocrit.

On kidney biopsy, an MPGN lesion is seen in more than 80% of cases (Fig. 22.6). Light microscopy findings that point toward a cryoglobulinemic MPGN include a greater number of macrophages than seen in other forms of proliferative glomerulonephritis and the presence of intraluminal thrombi composed of precipitated cryoglobulins. On IF, there is diffuse IgM deposition in capillary loops, and EM may show subendothelial deposits that often have a characteristic “fingerprint” pattern of cryoprecipitates. MPGN with features of cryoglobulins should prompt evaluation for infections (especially HCV),

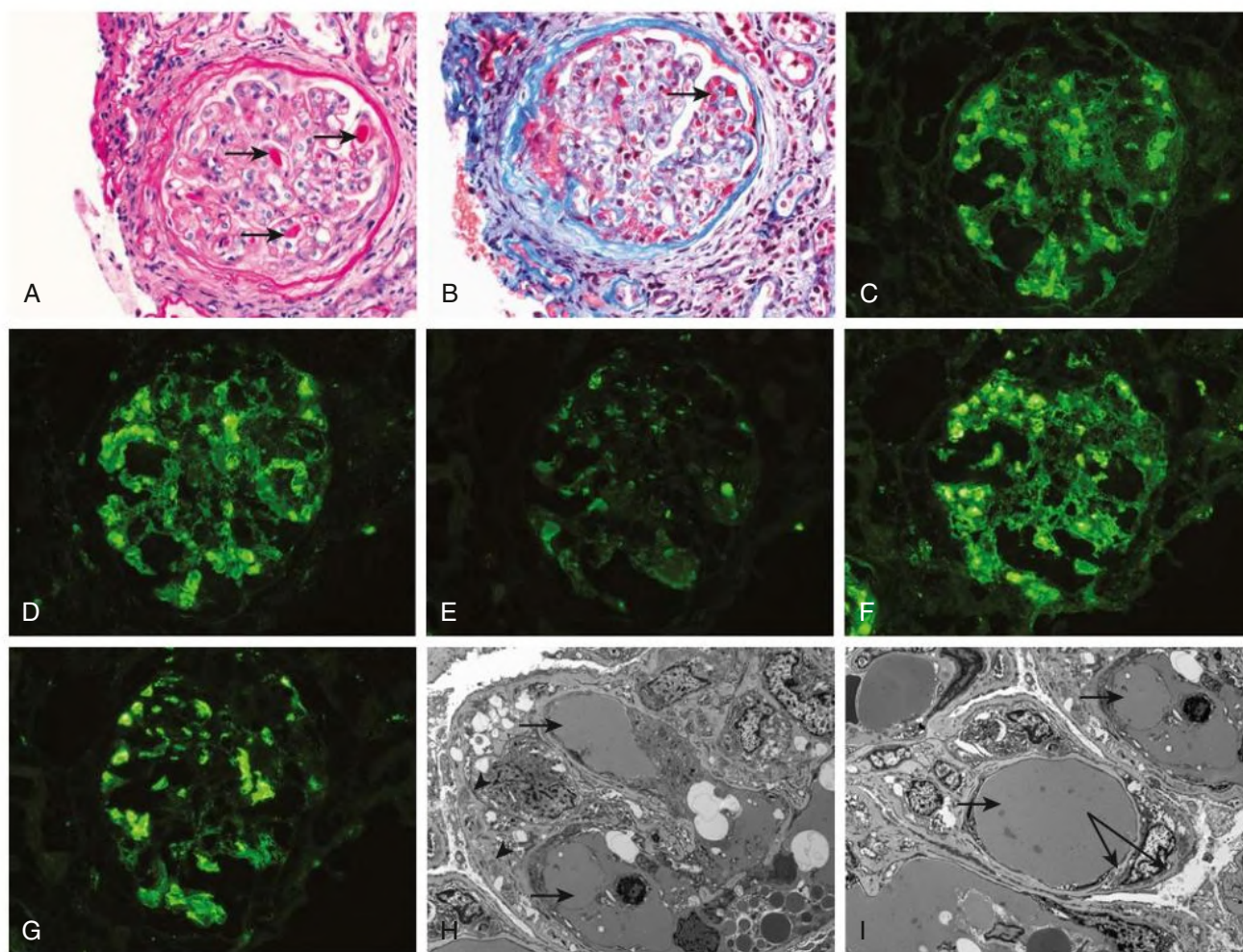


Fig. 22.6 Membranoproliferative Glomerulonephritis and Cryoglobulins Associated With Sjögren Syndrome. (A–B) Light microscopy showing a membranoproliferative glomerulonephritis pattern of injury. Immunofluorescence microscopy showing bright granular staining for immunoglobulin (Ig) G (C), IgM (D), C3 (E), kappa chains (F), and lambda light chains (G). (H–I) Electron microscopy showing numerous intraluminal deposits (arrowheads in H); subendothelial deposits and double contours (double arrow in I) are also present. Black arrows point to intraluminal deposits representing cryoglobulins. (A, PAS, $\times 40$. B, Masson trichrome stain, $\times 40$. H and I, $\times 2900$.)

autoimmune diseases (especially Sjögren syndrome), and monoclonal gammopathies. Features that suggest glomerulonephritis because of monoclonal immunoglobulin-associated cryoglobulins (type I cryoglobulins) include intraluminal periodic acid–Schiff (PAS) positive (hyaline-like) deposits on light microscopy, intraluminal monoclonal immunoglobulin on IF microscopy, and substructures (microtubules, fibrillary, fingerprints) on EM.

Treatment

Patients with mixed cryoglobulinemia who have mild disease (i.e., petechial rash without necrotizing lesions, mild sensory neuropathy, arthralgias) typically do not require immunosuppressive therapy, and management of such patients should focus on treating the underlying disease. Patients with a rapidly progressive, organ-threatening, or life-threatening cryoglobulinemic syndrome such as rapidly progressive glomerulonephritis, severe digital ischemia, gastrointestinal vasculitis, rapidly progressive neuropathy, central nervous system vasculitis, and heart failure should be treated with immunosuppressive therapy, regardless of the cause of the mixed cryoglobulinemia.⁴⁶ This typically involves a short course of

corticosteroids combined with rituximab. If rituximab is unavailable, fails to produce a response, or is not tolerated, cyclophosphamide can be used instead.

In patients with life-threatening disease (e.g., acute respiratory failure with pulmonary hemorrhage); cryoglobulinemia-associated hyperviscosity syndrome; severe, refractory skin ulcers because of cutaneous vasculitis; or a high cryocrit level (i.e., $\geq 10\%$), we suggest adding plasmapheresis to immunosuppressive therapy. We typically perform daily plasma exchange for 10 to 14 sessions or three exchanges per week for 2 to 3 weeks. After disease stabilization, therapy should be directed at the underlying disease. Patients with HCV infection (without decompensated cirrhosis) and mixed cryoglobulinemia should receive antiviral therapy as per current guidelines.⁴⁷

Patients with severe manifestations of cryoglobulinemic vasculitis associated with HCV infection and not yet treated with antiviral therapy should receive immunosuppression first, with antiviral therapy delayed for 1 to 4 months.⁴⁴ The rationale is based on the fact that immunosuppression can rapidly improve inflammation and resolve target-organ damage, although no randomized trials have evaluated immediate versus delayed timing of antiviral therapy. However, in

patients with cryoglobulinemic vasculitis associated with infection with HIV or HBV, antiviral therapy should always be initiated before or concurrently with immunosuppressive therapy because these patients are at high risk for enhanced viral replication as a result of rituximab or cyclophosphamide therapy.

The specific antiviral drugs and therapy schemes depend on the HCV genotype, kidney function, prior treatment response, type of

antiviral agent used, tolerance to treatment, and clinical and laboratory response to treatment.

Patients with a lymphoproliferative disorder should receive disease-specific therapy.

Patients treated with immunosuppressive therapy should receive opportunistic infection (i.e., *Pneumocystis jirovecii*) prophylaxis. The frequency of monitoring is determined by the severity of disease.

SELF-ASSESSMENT QUESTIONS

1. A 50-year-old woman is referred for sudden onset of edema and hypertension. Past medical history is unremarkable. Apart from a blood pressure of 155/95 mm Hg and 2+ edema, the physical examination is unremarkable. Serum creatinine is 1.8 mg/dL and proteinuria is 3.2 g/24 h. Urinalysis shows 50 to 100 red blood cells per high-power field, of which more than 25% are dysmorphic. C3/C4 complement levels, antineutrophil cytoplasmic antibody, antinuclear antibody, and anti-GBM are all normal or negative. A kidney biopsy is performed. Light microscopy shows an MPGN pattern of injury on immunofluorescence (IF) IgG (+++), IgM (+), IgA (–) C3 (+++), C1q (–), kappa (+++), lambda (–) chains, and EM duplication of the GBM with subendothelial deposits. The most likely diagnosis is:
 - A. MPGN secondary to monoclonal gammopathy
 - B. MPGN secondary to underlying infection
 - C. MPGN secondary to abnormalities in the alternative pathway of complement
 - D. MPGN type I (idiopathic)
2. A 65-year-old man undergoes orthotopic liver transplantation for liver failure secondary to HCV infection. The immediate posttransplantation course is uneventful. However, 1 year later he presents with recurrence of HCV infection, blood pressure of 150/90 mm Hg, proteinuria of 3 g/24 h, a rising serum creatinine up to 1.8 mg/dL, and a serum albumin of 3.9 g/dL. Urinalysis shows 40 to 50 red blood cells per high-power field, of which more than 25% are dysmorphic. C3 complement is 101 mg/dL (normal 75–115 mg/dL), C4 complement is 2 mg/dL (normal 10–40 mg/dL), and rheumatoid factor is positive. In this patient, a kidney biopsy is most likely to show:
 - A. Focal segmental glomerulosclerosis
 - B. Minimal change disease
 - C. Membranous nephropathy
 - D. Cryoglobulinemic glomerulonephritis
3. Which of the following causes of glomerulonephritis is typically associated with low serum complement C4?
 - A. Mixed cryoglobulinemia
 - B. Henoch-Schönlein purpura nephritis
 - C. Anti-GBM disease
 - D. Poststreptococcal glomerulonephritis
4. A 52-year-old woman with a long history of rheumatoid arthritis and Sjögren syndrome is evaluated for new onset of a skin rash on the lower extremities and impaired kidney function. Blood pressure is 150/90 mm Hg. Serum creatinine is 1.8 mg/dL. Urinalysis shows more than 50% red blood cells per high-power field, of which more than 25% are dysmorphic. Proteinuria is 2.8 g/24 h. C3 complement is 82 mg/dL (normal 75–115 mg/dL), and C4 complement is less than 5 mg/dL (normal 10–40 mg/dL). Antineutrophil cytoplasmic antibody, antinuclear antibody, and anti-GBM serologic results are all negative. In this patient a kidney biopsy is most likely to show:
 - A. MPGN with IgA (+++), C3 (++), IgG (+), kappa (+), lambda (++)
 - B. MPGN with linear IgG.
 - C. MPGN with C3 (+++), IgG (neg), IgM (neg), IgA (neg).
 - D. MPGN with IgG (++), IgM (+++), IgA (+), C3 (++), kappa (++), lambda (++)

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